

Ayoub ADIL

Software engineer, *with a 3D background.*

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I build software applications, starting with the architecture, and I keep looking after them once they're delivered. Part of my work is 3D, mostly point clouds and the viewers that display them. The rest is larger systems made of many services, plus the applications their operators use. Several of them run in production today.

EXPERIENCE

R&D Vision — 03/2024 – Present

Software Engineer · Same role, renamed when the company specialised its job titles — 09/2025 – Present

Application Engineer — 09/2024 – 09/2025

R&D Intern · Master's final internship: 3D reconstruction from images — 03/2024 – 08/2024

I've worked alone, in pairs and in bigger teams, depending on the project. I install what I deliver and support it afterwards.

INDUSTRIAL PROJECTS · 2024 – PRESENT

Tunnel inspection platform — In progress · 2026

Rail tunnel operator. Developed most of the services and the communication between them · Developed all the applications: operator tablet, multi-screen inspection, history comparison.

`.NET · WPF · C# · C++ · Modbus / JBus · Phoxene · NI DAQ`

LiDAR safety barrier for e-bus high-voltage charging — Proof of concept · 2026

Transport-sector manufacturer. Wrote the point-cloud processing that tells real intrusions from rain, snow and bus motion · Integrated and calibrated the LiDARs · Built the operator interface.

`Python · C++ · C# · WPF / .NET`

3D dough monitoring in an industrial oven — In production · 2024

Food-industry group. Integrated the ToF camera · Built the 3D viewer and the depth-image display · Built the operator interface. `C++`

Track inspection software, 3 vehicle configurations — In production · 2026

Rail inspection company. Took over a colleague's code and fixed malfunctions · Refactored it into one modular application for the 3 configurations · Installed it on trains · Support and maintenance. `.NET · WPF · C#`

Signal-box route-setting assistant — In production · 2025

Rail operator. Built the complete application · Recording, history and replay mode.

`.NET · WPF · C# · C++`

Fabric defect inspection — Delivered · 2025 – 2026

Luxury fashion house. Built the operator interface · Built the dataset-creation tools.

`C# · C++ · WPF / .NET`

Train passage detection — Delivered · 2026

Rail operator. Built the data-browsing application · Built the data service. `.NET · WPF`

Metro tunnel niche detection — In production · 2024

Urban transit operator. Took over a colleague's code · Finalised the niche-detection application.

`WPF`

MASTER'S INTERNSHIP & ACADEMIC PROJECTS

3D reconstruction of rail and metro tunnels from images · Master's internship · 03/2024 – 08/2024
Reviewed the visual-SLAM literature · Implemented and ran ORB-SLAM and PLVS · Applied them to real rail and metro footage and analysed the results. `C++ · Docker · Ubuntu`

Monte Carlo path tracing · Master course project · 12/2023 `C++ · OpenGL · Shaders`

Experimental protocols in virtual reality · Master project · ICube lab · 09/2023 – 12/2023
`Unity · VR`

Interaction with objects in virtual reality · Research project · ICube lab · 01/2023 – 06/2023
`Unreal Engine 5 · VR`

Smart home · Bachelor's internship · FST Mohammedia · 04/2020 – 06/2020 `Node.js · Python · TypeScript`

EDUCATION

2021 – 2024
Master's, Image & 3D
UFR Math-Info · University of Strasbourg

2020 – 2021
Bachelor's (final year), Computer Science
INP-ISIMA · Clermont Auvergne University

2019 – 2020
Bachelor's, Computer Science, Networks & Multimedia
FST Mohammedia · Hassan II University

2017 – 2019
DEUG: Mathematics, Computer Science & Physics
FST Mohammedia · Hassan II University

2016 – 2017
Baccalaureate, Mathematical Sciences
Lycée Ben Msik

SKILLS

LANGUAGES
C++, C#, Python, TypeScript, JavaScript

3D & GRAPHICS
OpenGL / shaders, three.js, Unity, Unreal Engine 5, Blender

VISION & DATA
OpenCV, Point clouds, Visual SLAM, Stereo calibration

FRAMEWORKS
.NET / WPF, React, Node.js

TOOLS
Git, Docker, Linux, Claude Code

HARDWARE
Line-scan cameras, Area-scan cameras, ToF cameras, Thermal cameras, LiDAR, NI DAQ, Modbus / JBus, RFID / GPS

LANGUAGES

Arabic	Native
French	Fluent
English	Fluent
German	Learning